

AI: Threat or Enabler?

Version: 1.2

Author: Sylvain Martinez

Classification: PUBLIC

{**elysium**security}
cyber protection & response

Welcome to this presentation on Artificial Intelligence and Cyber Security Risks. In this presentation, we will examine how AI is used by hackers and security professionals alike to attack and defend against cyber threats.

1. CONTEXT: EMERGING THREATS TODAY
2. THREAT: HOW AI IS WEAPONISED
3. ENABLER: HOW AI DEFENDS
4. NEXT: RECOMMENDATIONS & ROAD AHEAD



CONTEXT

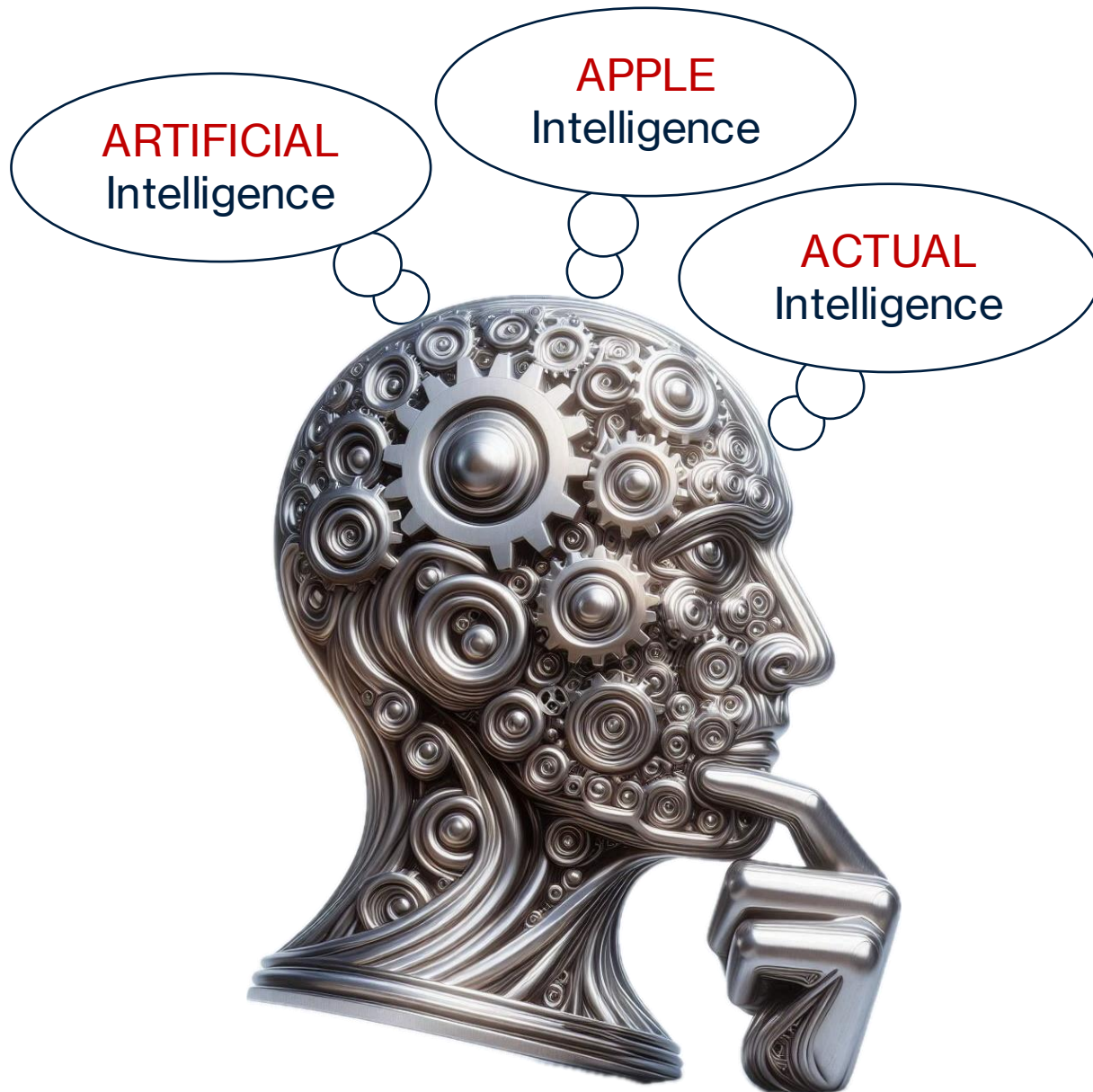
THREAT

ENABLER

NEXT

- IoT devices are a growing threat
- Ransomware attacks are becoming more sophisticated
- Nation-state attacks pose a major threat
- AI now embedded in both attack and defense toolkits





ARTIFICIAL Intelligence

Simulation of human intelligence in machines that are programmed to think, learn and work like humans.

TODAY

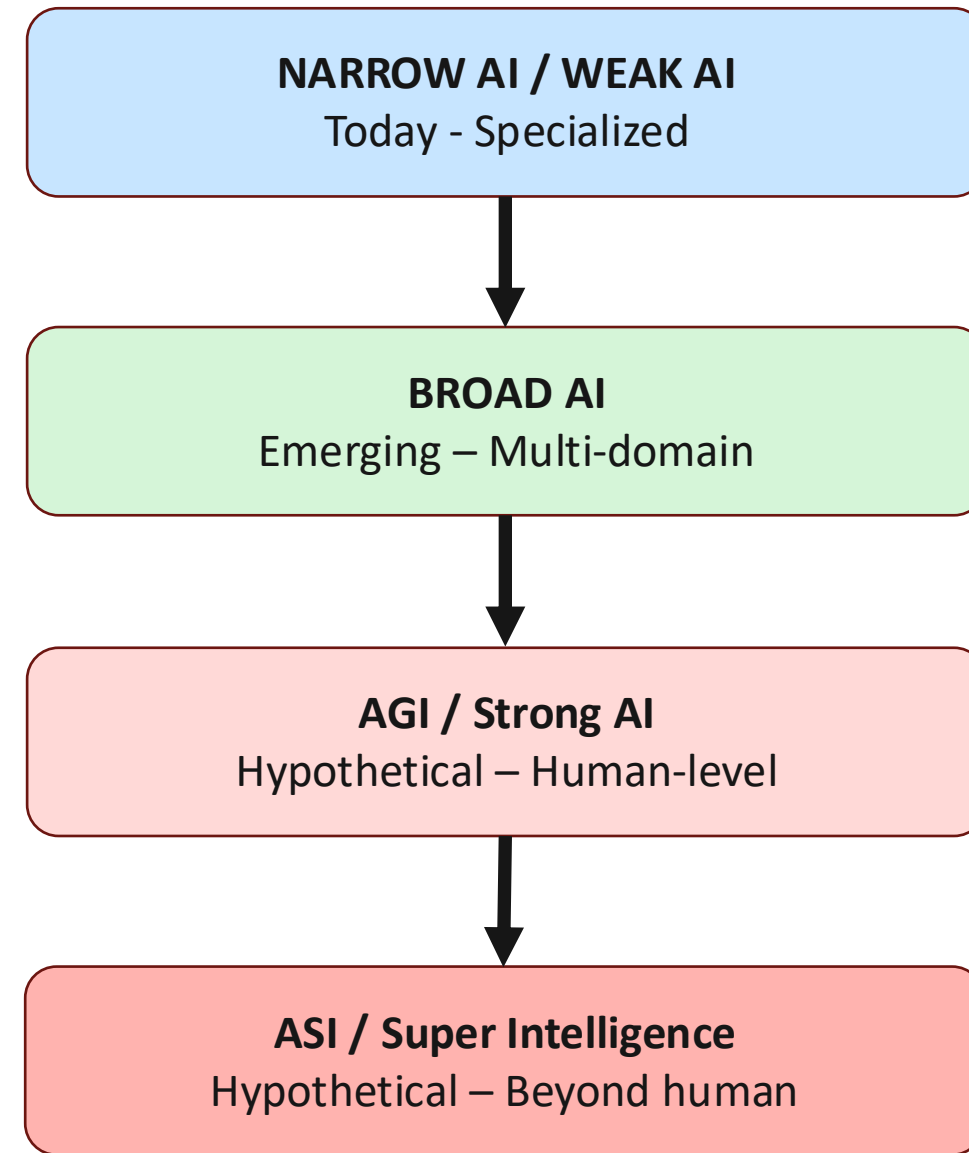
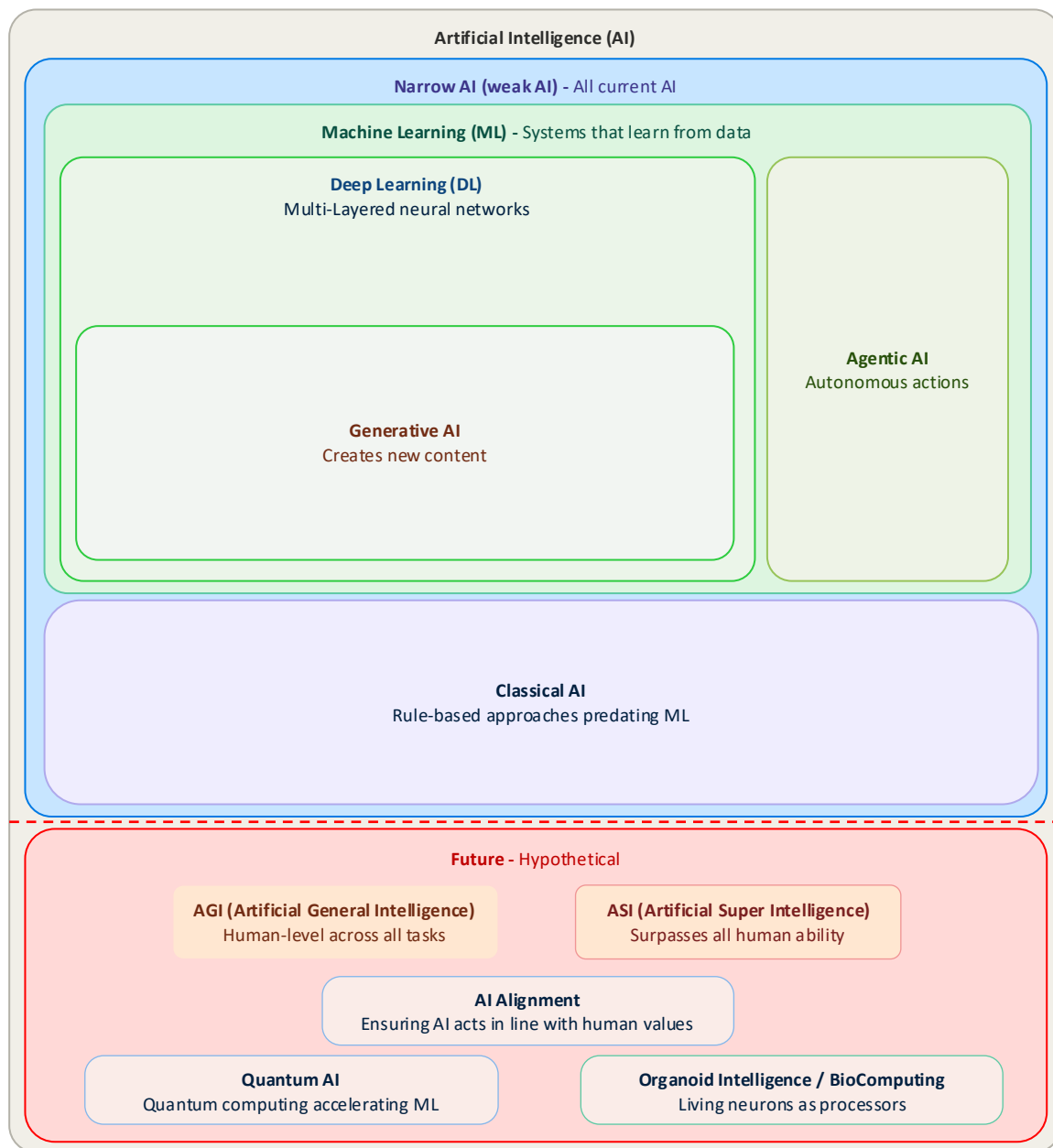
- NARROW AI: AI Designed for a specific task – All AI today
- MACHINE LEARNING (ML): Learns pattern from data
- DEEP LEARNING (DL): Learns complex large datasets
- NATURAL LANGUAGE PROCESSING (NLP): Human Language
- GENERATIVE AI: Creates new content
- LARGE LANGUAGE MODEL (LLM): Mass data training
- AGENTIC AI: Plans and takes actions

EMERGING

- AGENTIC WORKFLOW: Autonomous multi steps process

TOMORROW

- ARTIFICIAL GENERAL INTELLIGENCE (AGI): Human level
- ARTIFICIAL SUPER INTELLIGENCE (ASI): Post Human





DUAL USAGE

Used both for Attacks and Defense

Used both by Hackers/Security professionals

Both to enhance skills and speed



AUTOMATION

Used to automate attack

Used to automate threat detection



GUIDELINES

Practical guidelines are needed to ensure proper use of AI

In and Outside the organization
For Private and Public activities
For everyone – Laws/Standards



AI is currently an Add-on to cyber security practices
AI will become increasingly native to all cyber activities



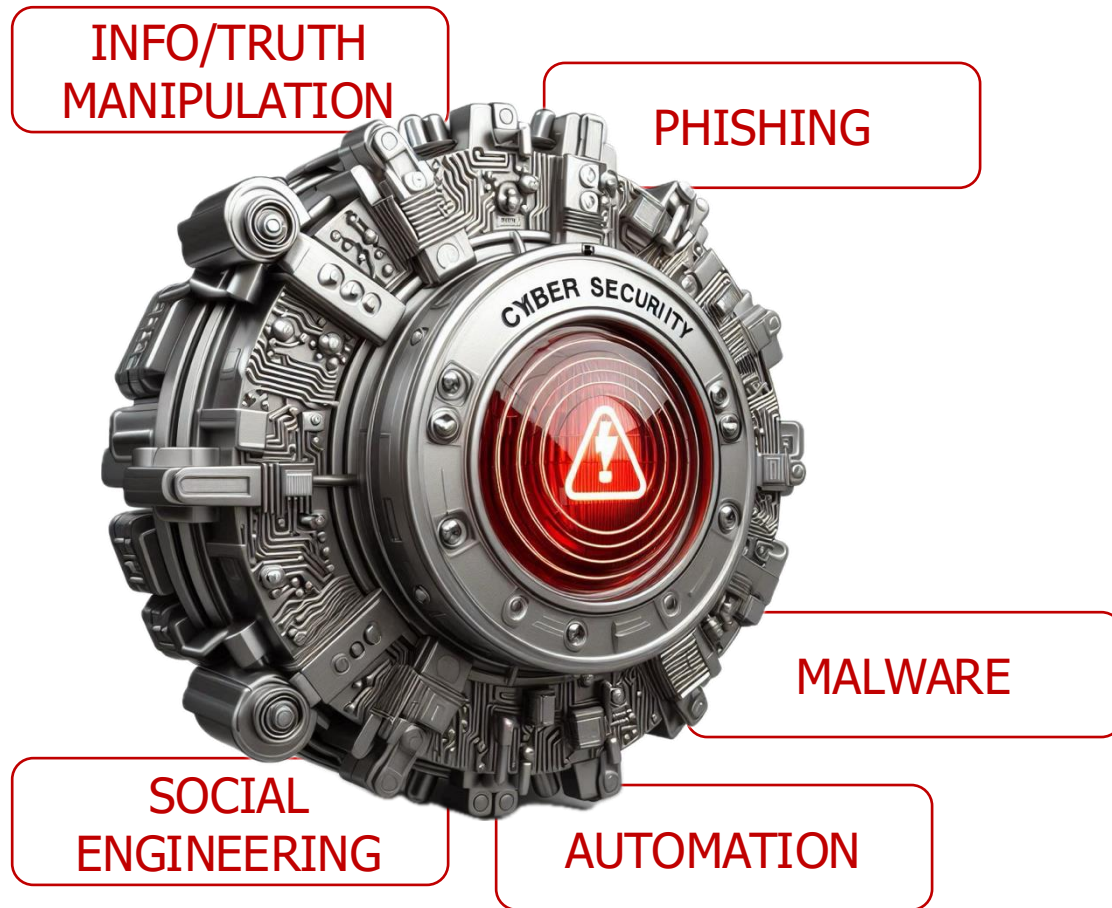
The diagram illustrates a four-step threat model process flow. It consists of four horizontal bars stacked vertically, each containing a step name. The bars are connected by a vertical line of four circles. The first, second, and fourth circles are grey, while the third circle is white. Red lines connect the top of each circle to the bottom of the circle above it, forming a continuous path through the steps.

CONTEXT

THREAT

ENABLER

NEXT



- **Enhanced:** AI assisted attacks are increasing in sophistication
- **Hacker Advantage:** AI attacks are more difficult to detect and prevent
- **Easy Access:** AI hacking tools allow low-skilled hackers to improve their hacking skills

QUICK FIND

AI is used to identify vulnerabilities in software, automatically, rapidly.

LOW SKILL ENTRY

Entry skill is mainly how to write a prompt, not necessarily deep and difficult knowledge to master

PERFORMANCE

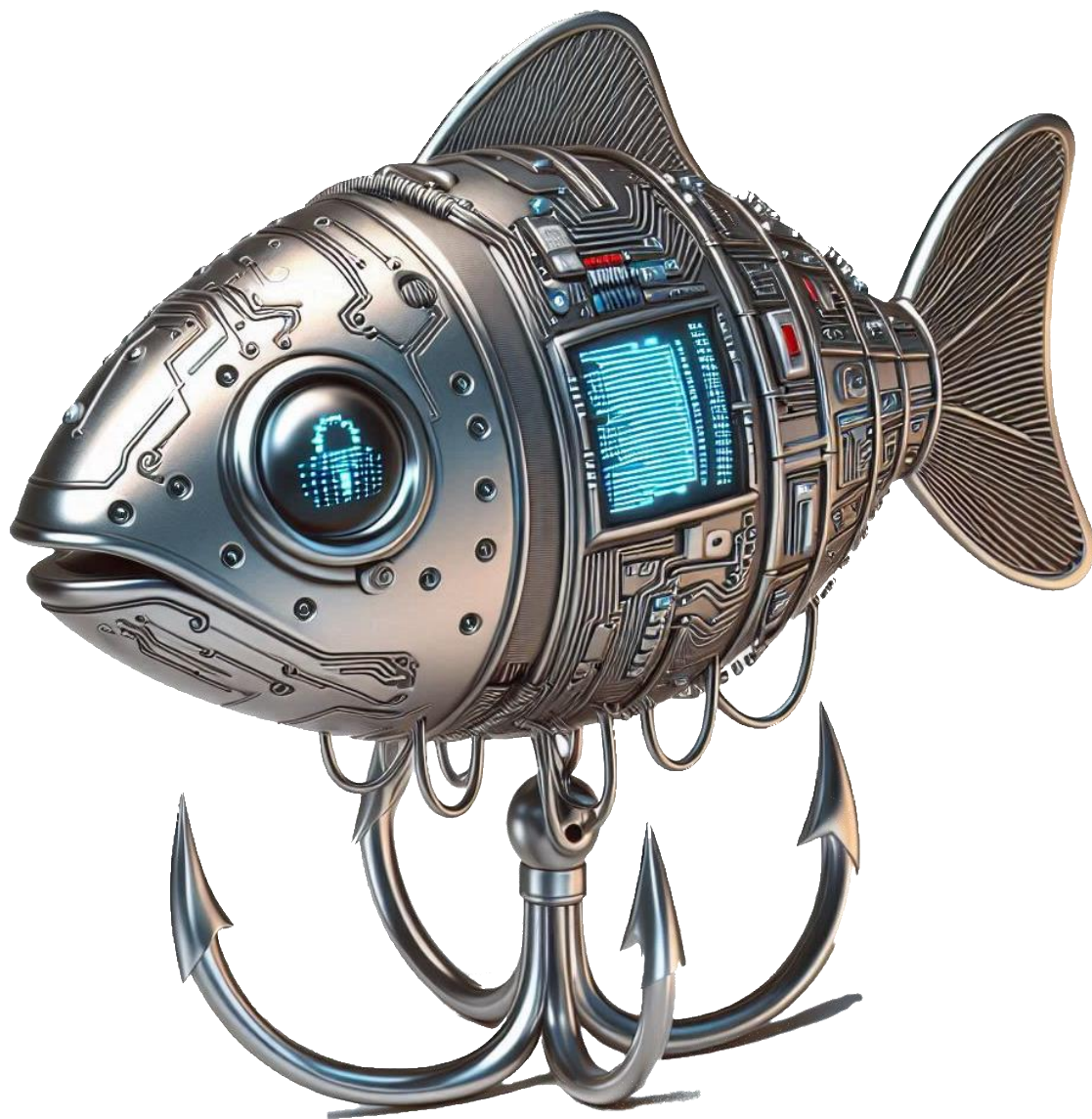
AI is used to evade detection by security software through **advanced** and **unusual** technics

CHAINED ATTACKS

AI can spread malware or enter a system in a more targeted manner, combining different steps and hacking technics:
Red Team NG



Moving towards a pentester on demand / as-a-service
or... a hacker on demand / as-a-service directly from a portal



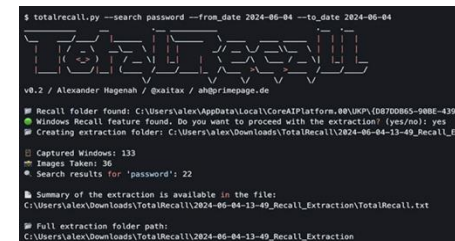
- **Realistic:** Hackers use AI to mimic trusted people to increase phishing attacks success
- **Harvesting:** AI is used to analyze data patterns, profile targets, and tailor phishing attacks
- **Interactive:** AI bots can provide live voice, video, or text responses to manipulate victims

WHERE TO FIND PERSONAL DATA?

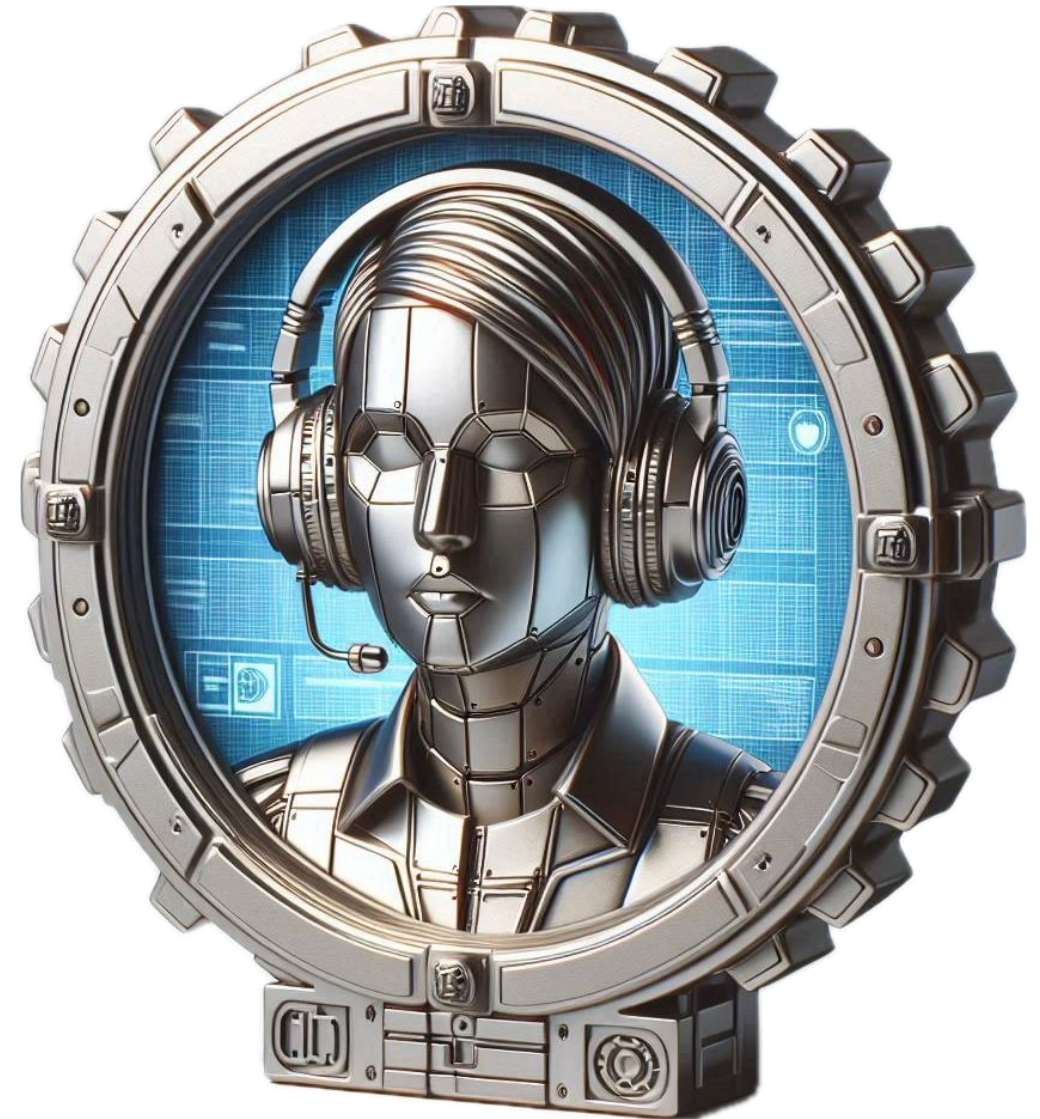
HACKERS GOLD?

APPLE
Intelligence

MS
Recall



- Deepfakes are **increasingly realistic** manipulated videos or audios content.
- They are made using **machine learning trained on real media**.
- The technology is becoming **more accessible**.
- Deepfake videos and audios can be used to **spread false information** or to defame individuals.
- Deepfake videos and audios can be **difficult to detect**.





LIMITED INPUT

Creating deepfake video and audio requires only limited source material.



LIVE GENERATION

Law enforcement dismantled major groups, Deepfake video and audio can now be generated in real time.



ACCESSIBLE TOOLS

Online tools are making deepfake creation easier and more accessible. (i.e.: Grok imagine)

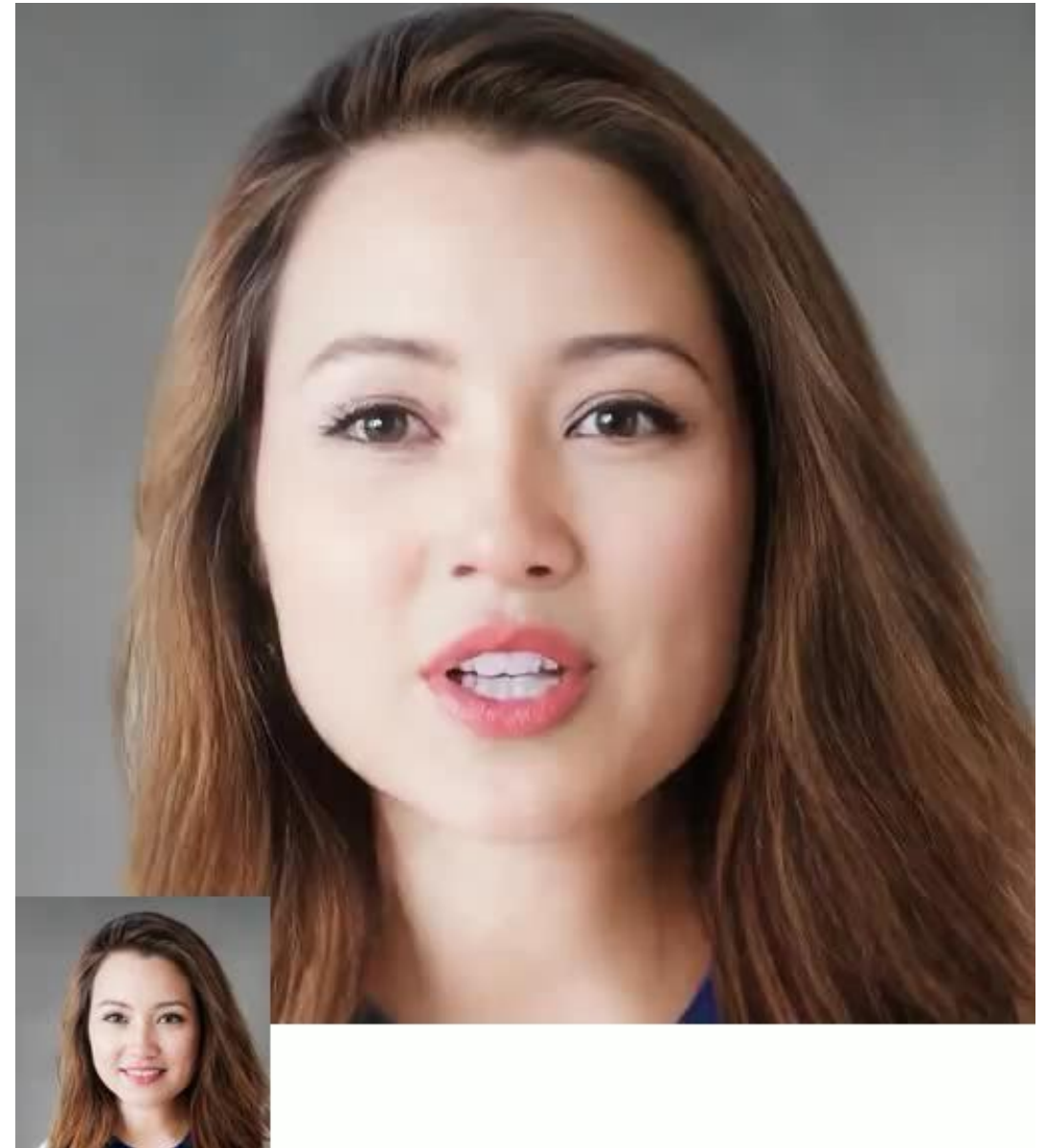


LOW BARRIER

Creating convincing deepfakes requires less data and technical skill.

- <https://www.microsoft.com/en-us/research/project/vasa-1/>
- April 2024
- VASA-1: Lifelike Audio-Driven Talking Faces Generated in Real Time
- Only requires a single static image and a speech audio clip
- Can produce synchronized lip movements, facial expression and emotions, head movements, etc.
- in real time...

- 1x photo = 1x video!



- 1x photo = Multiple videos!



- 1x photo = **Realtime** videos! (2D images generated to simulate 3D movement)

• Upload image, or pick one below

Upload



• Upload audio, record audio, or generate by TTS

Upload Generate Record

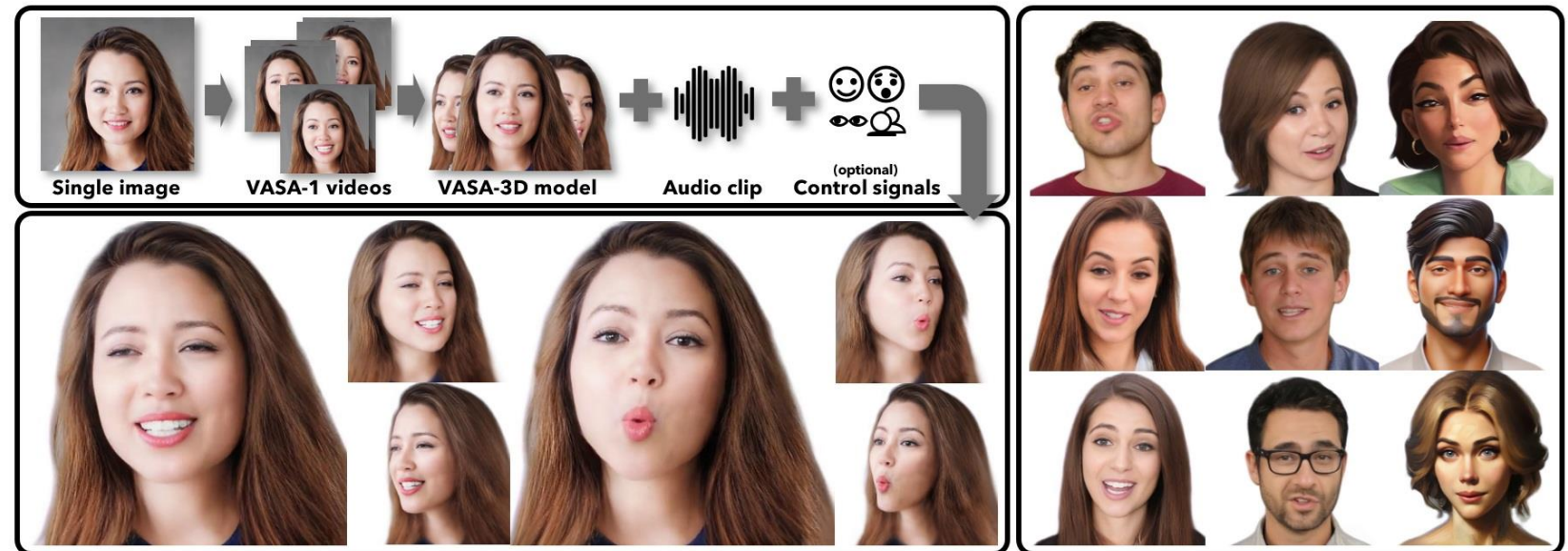
Talking face video live stream



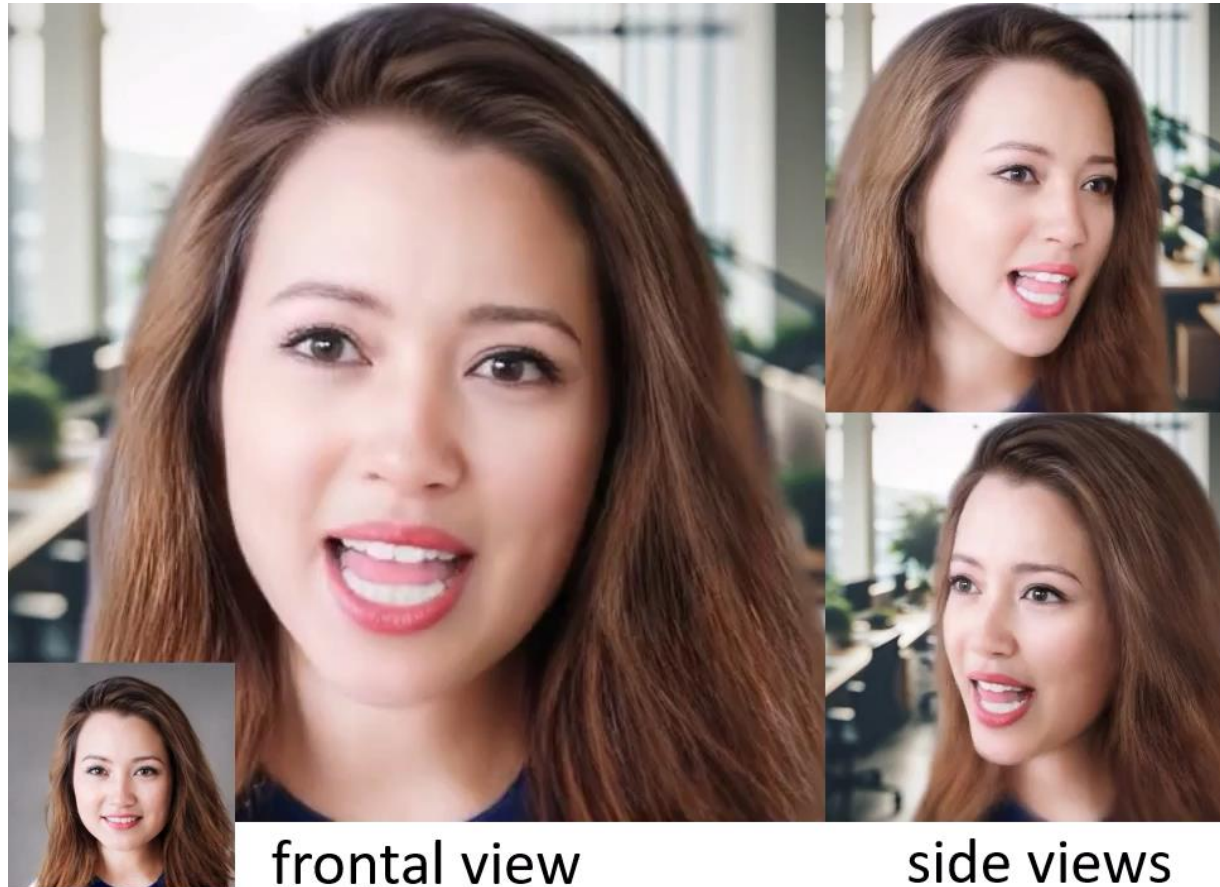
Pitch: 0.00
Yaw: 0.00
Roll: 0.00
X: 0.00
Y: 0.00
Z: 1.00
Gaze X: 0.00
Gaze Y: 0.00

Reset

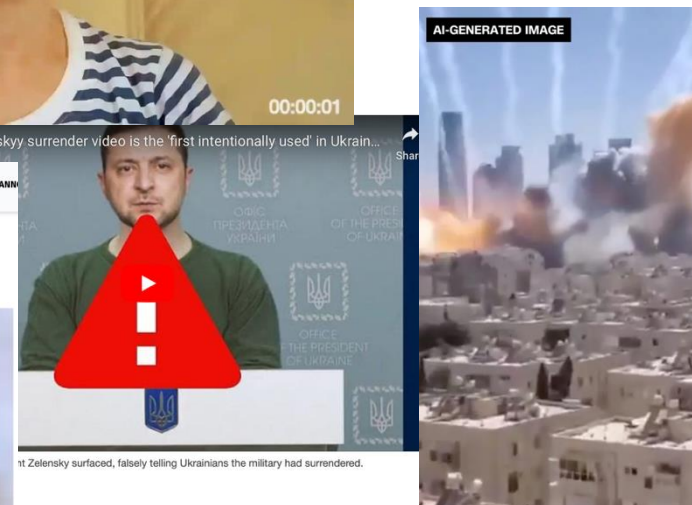
- <https://www.microsoft.com/en-us/research/project/vasa-3d/>
- October 2025
- VASA-3D: Lifelike Audio-Driven Talking Faces Generated in Real Time **as a 3D model**
- Only requires a single static image and a speech audio clip
- Can produce **precise lip-audio sync** movements, facial expression and emotions, head movements, etc.
- in real time...



- Desktop PC with a single NVIDIA RTX 4090 CPU – 512x512 size at 75fps



- 2024: a Hong Kong financial institution lost \$26 million due to a fake video conference with deepfake employees
- 2024: Various Mauritian CEO were providing fake financial advice
- Feb 2025: Italian Defence minister fake voice used for 1 million Euro business elite scam (free journalists)
- Sept 2025: Deepfake videos provided as evidence in Mendones v. Cushman & Wakefield
- 2026: Fake videos of war related news
- Mauritius 2024: Missie Moustass, Real or fake?





- **FIMI**: Foreign Information Manipulation & Interference
- **Scale**: AI can generate thousands of coordinated fake personas at scale
- **Poisoning**: LLM training data can be poisoned to introduce bias or hidden backdoors
- **Manipulation**: AI can poison public discourse by amplifying division and suppressing truth
- **Feedback loop**: Both a technical and human problem - systems and minds are targeted in a self-reinforcing loop

- **Evidence:** Deepfakes make video and audio evidence less reliable
- **Text:** AI-generated text is becoming harder to distinguish from human writing
- **Fabrication:** AI enables news fabrication at massive scale
- **Denial:** The “liar’s dividend” allows real evidence to be dismissed as AI-generated
- **Erosion:** Public trust in digital communications is steadily eroding

PLAUSIBLE DENIABILITY

OPINION MANIPULATION

DIGITAL OBSCURANTISM

WHEN EVERYTHING CAN BE FAKED, EVERYTHING CAN BE DENIED

PRE-PHOTOGRAPHY

Centuries of uncertainty.
Rumors, propaganda, painting
as “truth”

THE BRIEF WINDOW

About 150 years where photos,
video & audio were hard to fake.
A golden age of trust

THE DEEPFAKE AGE

AI makes fabrication trivial.
Trust collapses.
Back to square one?

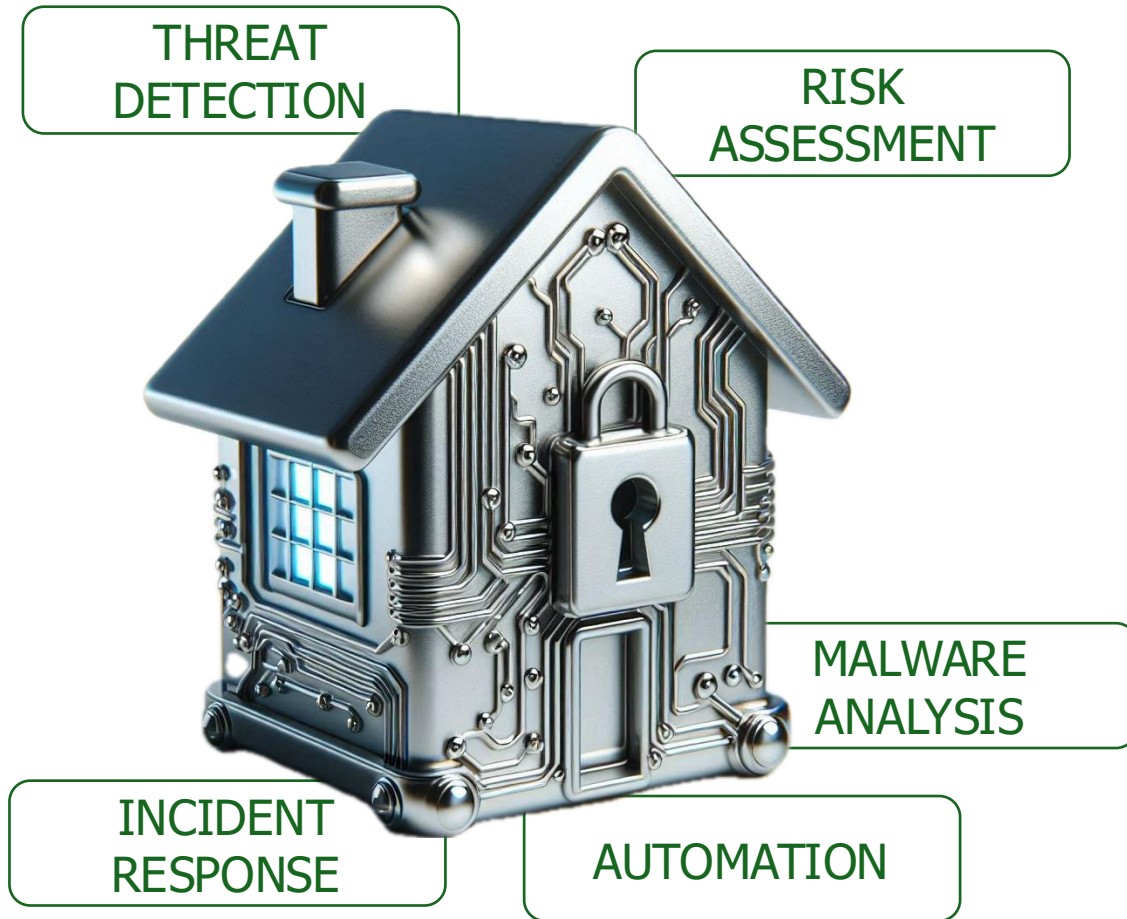


CONTEXT

THREAT

ENABLER

NEXT

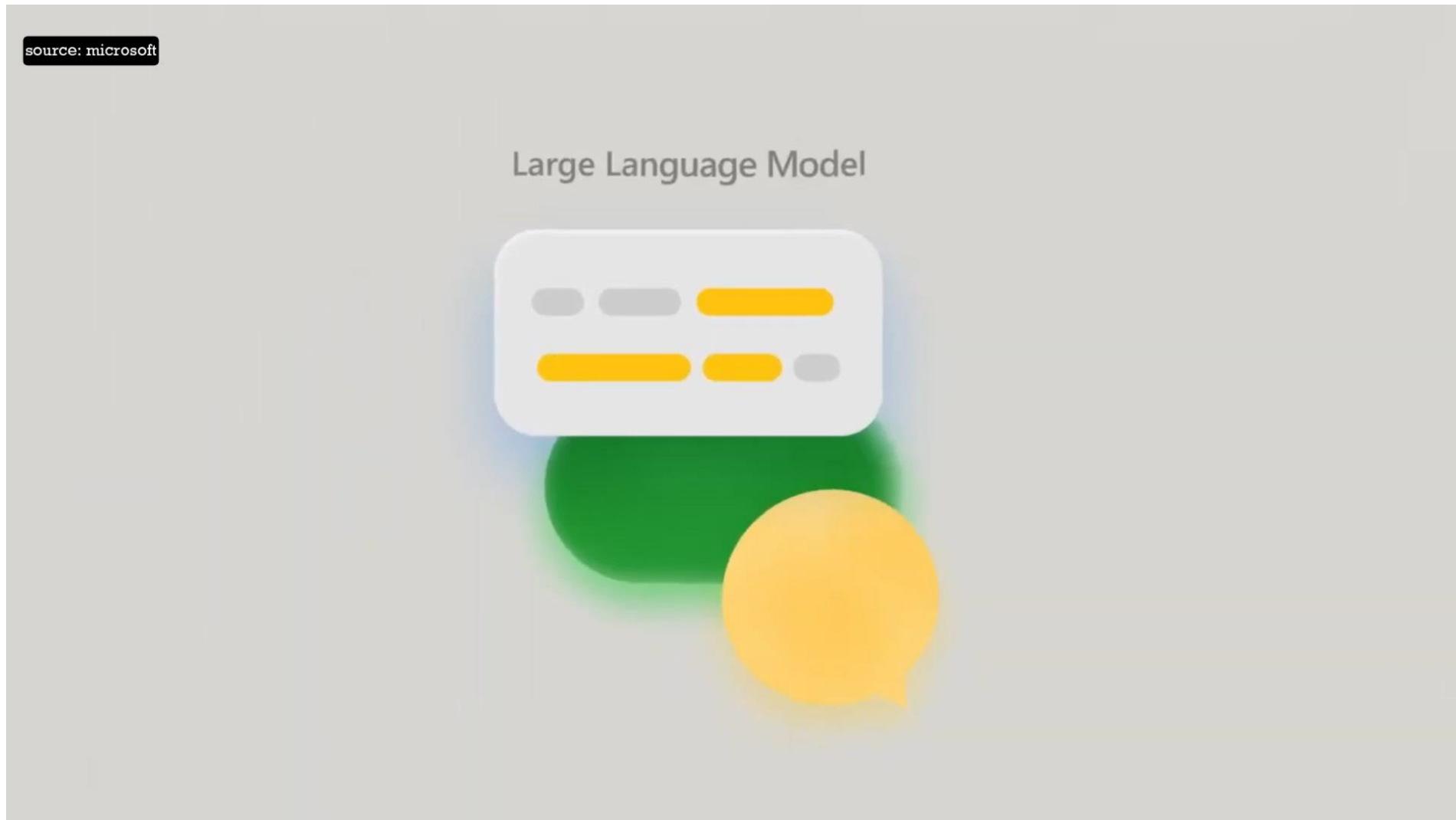


- **Maturity:** AI-assisted security tools are not entirely new
- **Speed:** AI security tools can accelerate complex analysis
- **Capacity:** AI security tools can help address shortages of **low to mid-level** analysts
- **Adaptation:** Organizations and security providers are starting to adapt

- **Correlation**: AI connects signals across billions of daily log events in SIEM platforms
- **Behaviour**: UEBA detects insider threats and compromised accounts through behavioural analysis
- **Anomaly detection**: AI can identify unusual activity that may indicate emerging threats
- **Precision**: AI reduces false positives so analysts can focus on real incidents
- **Speed**: AI-assisted monitoring can reduce time to detect from days to minutes



- Microsoft Security Copilot for Incident Response Analysis



- **Triage:** AI can rapidly assess and classify alerts and incidents
- **Prioritization:** AI can rank incidents based on risk, impact, and urgency
- **Guidance:** AI can recommend next steps, containment actions, and response playbooks
- **Automation:** AI can execute response actions and workflows automatically
- **Analysis:** AI can accelerate malware analysis and reverse engineering





- **Insights:** AI can analyze multiple risk factors to provide deeper insight into potential risks
- **Fraud:** AI can support fraud detection by identifying suspicious patterns and anomalies
- **Scale:** AI can accelerate due diligence by reviewing large volumes of data
- **Compliance:** AI can help monitor regulatory compliance and detect potential gaps
- **Accuracy:** Machine learning can improve the speed, consistency, and accuracy of risk assessment

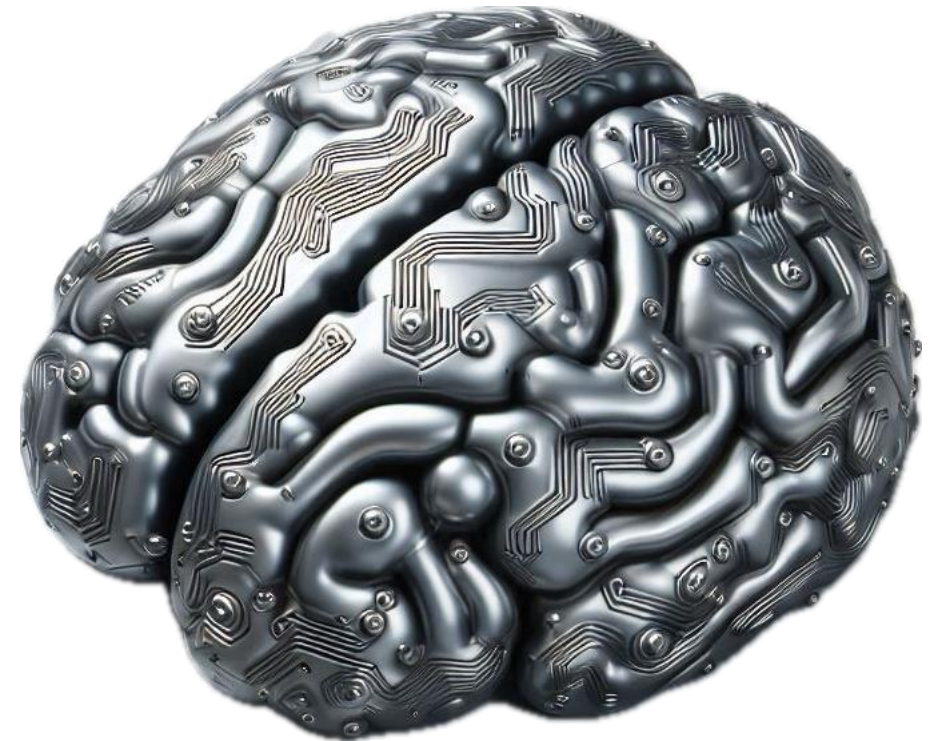
- **Recon**: AI can automate attack surface mapping in minutes
- **Emulation**: AI can simulate advanced adversary TTPs at far lower cost
- **Automation**: AI can accelerate repetitive testing tasks across large environments
- **Reporting**: AI can speed up documentation of findings and evidence
- **Coverage**: Continuous automated testing improves coverage beyond point-in-time assessments





- **Detection**: AI can be used to detect AI-driven attacks
- **Training**: AI can train models to identify AI-driven attacks more accurately
- **Defence**: AI can be used to build adaptive defensive systems
- **Response**: AI can help automate containment and response to AI-driven attacks
- **Integration**: AI-powered defences are already being integrated into major endpoint protection platforms

- **Awareness:** AI makes mistakes — learn to spot the signs
- **Verification:** Take the time to verify the identity of your interlocutor and the AI responses!
- **Scepticism:** The same old rule applies: if it is too good to be true...
- **Controls:** Basic cyber hygiene and strong security controls still matter





CONTEXT

THREAT

ENABLER

NEXT



DISRUPTION

Some criminal niches are under pressure as AI improves detection of common attack patterns



SCALE

Commodity cybercrime grows as AI amplifies ransomware-as-a-service and other mass-market attacks



EMPOWERMENT

Low-skill attackers are empowered — not replaced — by AI tools



ENHANCEMENT

Nation-state actors are enhanced, not replaced — AI is another operational weapon



AI makes more attackers more dangerous, not fewer



AUTOMATION

Tier 1 SOC work is increasingly automated, much of alert triage needs less human effort



OPERATIONS

AI can handle routine monitoring, compliance checks, and patch management tasks



EMPOWERMENT

Senior defenders can focus more on strategy and complex decisions



NEW SKILLS

New capabilities are required, including AI oversight, prompt engineering, and model governance



Fewer “today” jobs, but more impactful and better-compensated roles (Hopefully!)



- **Enterprises:** AI security tools are now a strategic investment, not an option
- **SMEs:** AI-powered attacks can target all organizations, but defences are not equally mature
- **Leadership:** AI risk must be treated alongside financial, legal & operational risk
- **Public sector:** Critical infrastructure faces growing AI-enhanced nation-state threats
- **Citizens:** Deepfakes, fraud, and manipulation affect the public, not just corporations

- **Expansion:** AI will continue to play a major role in cybersecurity
- **Innovation:** New AI-based tools and techniques will continue to emerge
- **Transformation:** Future cybersecurity systems will be designed with AI at their core





- **Awareness:** Understand AI capabilities and limitations
- **Use cases:** Define clear and appropriate use cases
- **Data:** Ensure data quality, scope, and integrity
- **Performance:** Assess AI model performance regularly
- **Confidentiality:** Beware of confidential information leakage
- **Positioning:** Decide where to lead, where to adopt, and where to wait

Be part of the solution

Engage with the stakeholders

Facilitate discussions, do not only say no

Be open minded (which is different from YES MAN)

Learn in earnest

New skillset needed – on top of basics

Be curious again – continuous improvement for job survival, not just promotion!

Differentiate hype versus real advances

Brace for impact

Awareness / DR / IR plans adapted to AI

Do not neglect baseline security

Ability to unplug AI and still run your business



Do not fight AI or be blindly excited about it
Embrace AI and be cautiously excited about it

THANK YOU!

Any Questions?



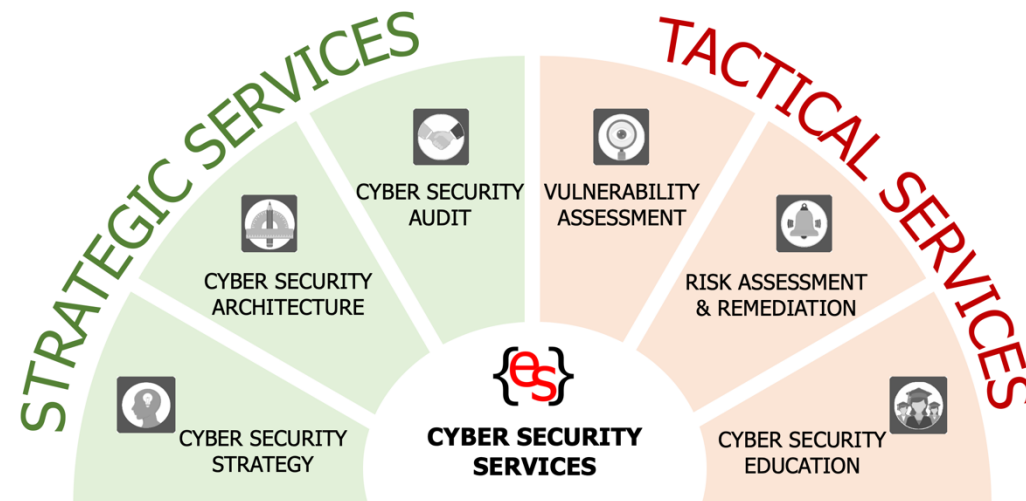
ABOUT ELYSIUMSECURITY LTD.

ELYSIUMSECURITY PROVIDES A PORTFOLIO OF **STRATEGIC** AND **TACTICAL SERVICES** TO HELP COMPANIES **PROTECT** AND **RESPOND** AGAINST **CYBER SECURITY THREATS**. WE DIFFERENTIATE OURSELVES BY OFFERING **DISCREET, TAILORED** AND **SPECIALIZED** ENGAGEMENTS.

ELYSIUMSECURITY OPERATES IN **MAURITIUS** AND IN **EUROPE**, A **BOUTIQUE STYLE** APPROACH MEANS WE CAN **EASILY ADAPT** TO YOUR BUSINESS OPERATIONAL MODEL AND REQUIREMENTS TO PROVIDE A **PERSONALIZED SERVICE** THAT FITS YOUR WORKING ENVIRONMENT.

ELYSIUMSECURITY PROVIDES **HIGH LEVEL EXPERTISE** GATHERED THROUGH YEARS OF **BEST PRACTICES EXPERIENCE** IN LARGE INTERNATIONAL COMPANIES ALLOWING US TO PROVIDE ADVICE BEST SUITED TO YOUR **BUSINESS OPERATIONAL MODEL** AND **PRIORITIES**.

ELYSIUMSECURITY PROVIDES **PRACTICAL EXPERTISE** TO **IDENTIFY VULNERABILITIES**, ASSESS THEIR **RISKS** AND **IMPACT, REMEDIATE** THOSE RISKS, **PREPARE** AND **RESPOND** TO INCIDENTS AS WELL AS RAISE **SECURITY AWARENESS** THROUGH AN ORGANIZATION.



contact@elysiumsecurity.com

<https://www.elysiumsecurity.com>